

M5: Homework

● Graded

Student

Jonathan C Elder

Total Points

42.35 / 50 pts

Question 1

5.1.2

3.5 / 5 pts

✓ - 0.5 pts c partially correct

✓ - 0.5 pts b partially correct

✓ - 0.5 pts a partially correct

Question 2

5.2.2

5 / 5 pts

✓ - 0 pts Correct

Question 3

5.3.2

5 / 5 pts

✓ - 0 pts Correct

Question 4

5.4.3

5 / 5 pts

✓ - 0 pts Correct

Question 5

5.5.1

4.5 / 5 pts

✓ - 0.5 pts c partially correct

Question 6

5.6.2

5 / 5 pts

✓ - 0 pts Correct

Question 7

5.7.2

5 / 5 pts

✓ - 0 pts Correct

Question 8

5.9.2

3.75 / 5 pts

✓ - 1.25 pts d incorrect

Question 9

5.10.5

3 / 5 pts

✓ - 1 pt a partially correct

✓ - 1 pt b partially correct

Question 10

5.12.2

2.6 / 5 pts

✓ - 0.6 pts a partially correct

✓ - 0.6 pts b partially correct

✓ - 0.6 pts c partially correct

✓ - 0.6 pts d partially correct

Question assigned to the following page: [1](#)

HW 5.1

5.1.2 a) Total Characters = Digits + Letters + Special Characters
 $= 10 + 26 + 4$
 $= 40$

Number of Passwords = 40 down till here are 6 characters
 $= 40(39)(38)(37)(36)(35)$
 $= 2,763,633,600$

b) total char = 40

$$P_7 = 40(39)(38)(37)(36)(35)(34) = 939,635,424,000$$

$$P_8 = 40(39)(38)(37)(36)(35)(34)(33) = 3,109,796,899,200$$

$$P_9 = 40(39)(38)(37)(36)(35)(34)(33)(32) = 992,255,000,774,400$$

$$P_7 + P_8 + P_9 = 1,024,202,612,160,000$$

c) first character = total (40) - Letters (26)
 $= 14$

$$P_7 = 14(39)(38)(37)(36)(35)(34) = 32,887,239,840$$

$$P_8 = 14(39)(38)(37)(36)(35)(34)(33) = 1,085,278,914,720$$

$$P_9 = 14(39)(38)(37)(36)(35)(34)(33)(32) = 34,728,925,271,040$$

$$P_7 + P_8 + P_9 = 35,847,091,425,600$$

Question assigned to the following page: [2](#)

5.2 HW

5.2.2 a) Show a bijection between P_6 and B^3

P_6 = palindromes of length 6.

B^3 = bit strings of length 3.

By definition: length: $P_6 \rightarrow B^3$

Let $x, y \exists x \exists y \in P_6$ Length($abc cba$) = abc

Ex. Length(101101) = 101 where $a, b, c \in \{0, 1\}$

Length(001100) = 001

Let Length($a_1 b_1 c_1 c_1 b_1 a_1$) = Length($a_2 b_2 c_2 c_2 b_2 a_2$)

$$a_1 b_1 c_1 = a_2 b_2 c_2$$

$$a_1 = a_2 \quad b_1 = b_2 \quad c_1 = c_2$$

length is one to one and length is a bijection.

b) What is $|P_6|$?

$$|P_6| = |B^3|$$

B has 3 places, with each place having 2 options $\{0, 1\}$

$$|B^3| = 2^3 = 8 \quad |P_6| = 8$$

c) Define: $F: P_7 \rightarrow B^4$ as

$$F(abcddcba) = abcd$$

$$\text{Ex. } F(00011000) = 0001$$

F is a bijection so $|P_7| = |B^4|$

$B = \{0, 1\}$ - 2 options w/ 4 places

$$|B^4| = 2^4 = 16$$

$$|P_7| = 16$$

Question assigned to the following page: [3](#)

5.3 HW

5.3.2 set $\{a, b, c\}$ with length 10. No consecutive characters.

First character has 3 choices $\{a, b, c\}$

2nd-10 has 2 choices. (can not be previous character)

$$\begin{aligned}\text{Number of strings} &= 3 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \\ &= 3 \times 2^9 \\ &= 1536\end{aligned}$$

Question assigned to the following page: [4](#)

5.4 HW

5.3 a) 10 people - no constraints

$$10! = 3628800$$

b) P and VP act as 1 member with 2 interchangeable options.
In this way there are 9 members with twice
as many permutations

$$= 2 \times 9!$$

$$= 725760$$

c) President and secretary act as 1 member with 2 options.
VP and treasurer act as 1 member with 2 options.

$$= 2 \times 2 \times 8!$$

$$= 161,280$$

Question assigned to the following page: [5](#)

5.5 HW

S.S.1 a) What is the value of f on input $(12, 1, 3, 15, 9)$?

Number of arrangements: $5!$
Number of permutations: $5!$

$$\frac{5!}{5} = 4!$$

Number of permutations $\{12, 1, 3, 15, 9\} = 24$

b) is $(12, 3, 12, 4, 19)$ a 5-permutation?

It is not a 5-permutation because the element 12 is repeated.

c) How many permutations are mapped onto the subset $\{12, 3, 13, 4, 19\}$?

$$\text{Number of permutation} = \frac{5!}{5(5-5)!} = \frac{5!}{5} = 24$$

Question assigned to the following page: [6](#)

5.6 HW

5.6.2

- a) 120 = Pianists
30/120 past 1st round
 $P(120, 30)$ or $P(120, 30)$

$$= \frac{120!}{30!(120-30)!} = \frac{120!}{30!(90)!} = \boxed{1.6974539e^{28}}$$

- b) 30 remaining pianists
5 total placings that are distinct

$$P(30, 5) = 30 \times 29 \times 28 \times 27 \times 26 \\ = \boxed{17100720 \text{ ways}}$$

Question assigned to the following page: [7](#)

S.7 HW

S.7.2 a) total # of cards in deck : 52
total # of distinct clubs : 13

$$52 \text{ cards} = C(n, r) = \frac{n!}{(r!(n-r)!)} \\ n = 52 \\ r = 5$$

$$= \frac{52!}{5!(52-5)!}$$

$$= \frac{52!}{5!(47!)} = 2598960 \text{ total / number of 5 card sets.}$$

without a club: $52 - 13 = 39$

$$C_2(n, r) = \frac{39!}{5!(39-5)!} = 575757$$

$$\text{set of 5 cards with a club} = 2598960 - 575757 \\ = 2023203$$

b) Number of ranks = 13

5 card hands with 13 rank combinations.

$$C(n, r) = \frac{13!}{5!(13-5)!} = \frac{13!}{5!(8)!} = 1287$$

possible ways to sort 4 suits with 5 cards = 4^5

$$= 1287 \times 4^5$$

$$= 1287 \times 1024$$

$$= 1317888$$

$\therefore$ Number of 5 card sets with at least 2 cards of the same rank = $2598960 - 1317888$

$$= 1281072$$

Question assigned to the following page: [8](#)

5.9 HW

5.9.2 a) total number of ways $\left[\begin{matrix} 15(b-1) \\ 15 \end{matrix} \right] \rightarrow {}^{20}C_{15} = 15504 \text{ ways}$

b) 3 must be chocolate

$$\text{Total}(15) - 3$$

$$\left[\begin{matrix} 12+(b-1) \\ b-1 \end{matrix} \right] = {}^{17}C_5$$

$$\sum_{k=3}^{15} \binom{19-k}{4} = 15504 - \sum_{k=0}^2 \binom{19-k}{4} = 6188 \text{ ways}$$

c) if only 2 can be sugar.

$$\text{total \# of ways} = 15504 - 6188 \text{ ways}$$
$$\text{remaining} = 9316$$

d)

Question assigned to the following page: [9](#)

5.10 HW

5.10.5 a) total different ways = ${}^{25}C_{10}$ m C n

$$= \frac{25!}{10!(25-10)!}$$

$$= \frac{25!}{10!(15)!}$$

$$= 3268760$$

b) total ways = $P(m, n)$

$$= \frac{25!}{(25-10)!}$$

$$= \frac{25!}{15!} = 11,861,676,288,000$$

Question assigned to the following page: [10](#)

5.12 HW

12.2 a) $n=1$ "No restrictions"
 $r=6, n=25$
 $(25-1)C_{(6-1)} = 24C_5$

$$= \frac{24!}{5!(24-5)!}$$

$$= \frac{24!}{5!(19)!} = \boxed{42504}$$

b) $25-18=7$

$$(7-1)C_{(6-1)} = 6C_5 = \frac{6!}{5!(6-5)!}$$

$$= \frac{6!}{5!(1)!} = \boxed{6}$$

c) $x \leq 3 = 25-3$

$$= 22$$

$$(22-1)C_{(6-1)} = 21C_5 = \frac{21!}{5!(16)!} = 20349$$

and

$$x \leq 10 = 25-10$$

$$= 15$$

$$(15-1)C_{(6-1)} = 14C_5 = 2002$$

$$21C_5 - 14C_5 = 20349 - 2002$$

$$= \boxed{18347}$$

d) $6(20-1)C_{20} + 6(6-1)C_6 - 6(12-1)C_{12} - 6(14-1)C_{14}$

$$= 25C_{20} + 11C_6 - 17C_{12} - 19C_{14}$$

$$= \frac{25!}{20!(5)!} + \frac{11!}{6!(5)!} - \frac{17!}{12!(5)!} - \frac{19!}{14!(5)!}$$

$$= 53130 + 462 - 6188 - 11628$$

$$= \boxed{35776}$$